

# Junran Wang

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## EDUCATION

|                |                                                                                                                                             |                                |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 2025 - Present | <b>Georgia Institute of Technology</b><br>Master of Science in <i>Electrical and Computer Engineering</i>                                   | Atlanta, GA<br>GPA: 4.0/4.0    |
| 2021 - 2025    | <b>Tsinghua University</b><br>Bachelor of Engineering in <i>Vehicle Engineering (Electronic Information)</i>                                | Beijing, China<br>GPA: 4.0/4.0 |
| 2024           | <b>University at Buffalo, the State University of New York</b><br>Visiting Scholar in <i>Department of Computer Science and Engineering</i> | Buffalo, NY                    |

## RESEARCH EXPERIENCE

**ImmersedPrivacy: Evaluating VLMs' Privacy Awareness in the Physical World** Jan 2026 - May 2026  
*Georgia Institute of Technology* | *First Author* | **NeurIPS 2026** |  [Website](#) Advisor: Dr. Pan Li

- Built a **reusable data generation and evaluation pipeline** in Unity for agent privacy, producing **400+ interactive audio-visual scenarios across 40+ physical environments** with configurable layouts and social contexts.
- Established evaluation quality through **human-annotated baselines** and controlled scenario augmentation, separating perception failures from privacy reasoning for **attribution analysis**.
- Benchmarked **12 VLMs** and exposed a **gap between privacy knowledge and behavior**: even the best model jointly satisfied task and privacy requirements in only **51%** of conflicting-command cases.

**Beyond Steering Vector: Flow-based Activation Steering** Mar 2026 - May 2026  
*Georgia Institute of Technology* | *First Author* | **NeurIPS 2026** |  [Website](#) Advisor: Dr. Chao Zhang

- Introduced FLAS, a **flow-based activation steering method** that learns concept-conditioned velocity fields to adapt interventions to each token's evolving activation state.
- Revealed **curved, multi-step, token-dependent steering** trajectories through activation-space analysis, providing empirical evidence that challenges the **linear representation hypothesis** in activation steering.
- Achieved **zero-shot generalization** to unseen concepts without per-concept tuning, becoming the first learned steering method to consistently outperform prompting on AxBench across Gemma-2-2B/9B.

**SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents** Jul 2026 - Sept 2026  
*Georgia Institute of Technology* | *First Author* |  [Website](#) Advisor: Dr. Pan Li

- Formulated agent attack and defense as a **POMDP** and implemented an **agent safety testing sandbox** with **resettable environments, trajectory replay, and executable checks** across filesystem, terminal, database, and web tasks.
- Developed DART, an **evolving attack agent** that refines strategies through execution feedback and tree search, using a **27B model** to attack commercial agents and achieving **73.3% attack success** against GPT-5.6.
- Developed SAGE, a **defender agent**, through **agentic post-training**, which actively **investigates environment** for safety evidence and reduces attack success to **at most 4.0%** under various adaptive attacks.

**AcFlow: Controlling T2I DiTs via Learned Conditional Activation Flow** Jul 2026 - Sept 2026  
*Georgia Institute of Technology* | *First Author* |  [Website](#)

- Developed ACFlow as a **scalable, text-conditioned, continuous-strength control module** for frozen image generators, enabling control toward a **general-purpose controller** for diffusion transformers.
- Scaled concept coverage without per-concept fitting: one controller covers **15,000+ styles** or **1,000+ concepts** and **generalizes** to unseen concepts.
- Validated the control interface on **FLUX and Z-Image**, achieving the **best intervention-preservation trade-off** among evaluated baselines.

## WORK EXPERIENCE

**Machine Learning Engineer Intern** Sept 2024 - May 2025  
*Infplane AI Ltd.* | *Pre-Series A* | *approx. CNY 50M raised*

- Ran small-scale **LLM pretraining experiments** with **Megatron-LM** on OpenWebText.
- Prepared pretraining corpora through **text cleaning, language/length filtering, and LSH-based deduplication**.
- Tokenized cleaned corpora and converted them into **Megatron-LM indexed binary datasets** for training.

PUBLICATIONS

*\* denotes equal contribution.*

[1] **How far are VLMs from privacy awareness in the physical world? an empirical study**  
**Junran Wang\***, Xinjie Shen\*, Zehao Jin\*, and Pan Li. **NeurIPS 2026** [Paper].

[2] **Beyond Steering Vector: Flow-based Activation Steering for Inference-Time Intervention**  
Zehao Jin\*, Ruixuan Deng\*, **Junran Wang\***, Xinjie Shen, and Chao Zhang. **NeurIPS 2026** [Paper].

[3] **Accelerated Sequential Flow Matching: A Bayesian Filtering Perspective**  
Yinan Huang, Hans Hao-Hsun Hsu, **Junran Wang**, Bo Dai, and Pan Li. **ReALM-GEN 2026 - ICLR 2026 Workshop (Spotlight)** [Paper].

[4] **SEAD: A State-Based Perspective on Attack and Defense in Tool-Using Agents**  
**Junran Wang\***, Xinjie Shen\*, Rongzhe Wei, and Pan Li. *arXiv preprint (2026)* [Paper].

[5] **AcFlow: Controlling Text-to-Image Diffusion Transformers via Learned Conditional Activation Flow**  
**Junran Wang\***, Zehao Jin\*, Tianyu Luan, and Xinjie Shen. *arXiv preprint (2026)* [Paper].

[6] **Transformers Stop Thinking Too Early, and a Tiny LoRA Fixes It**  
Zehao Jin, Ruixuan Deng, and **Junran Wang**. *arXiv preprint (2026)* [Paper].

HONORS & AWARDS

|                                                         |                                     |      |
|---------------------------------------------------------|-------------------------------------|------|
| National Scholarship                                    | Top 2% university students in China | 2024 |
| Overall Excellence Scholarship                          | Top 5% in Tsinghua University       | 2023 |
| Finalist in Interdisciplinary Contest in Modeling (ICM) | Top 2% teams worldwide              | 2023 |
| Overall Excellence Scholarship                          | Top 5% in Tsinghua University       | 2022 |

TECHNICAL SKILLS

|                       |                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------|
| Programming Languages | Python, C, C++; familiarity with Rust                                                                            |
| Training & Inference  | PyTorch, <b>CUDA</b> , <b>Diffusers</b> , <b>Transformers</b> , <b>Megatron-LM</b> , <b>vLLM</b> , <b>SGLang</b> |
| Agents & Evaluation   | <b>Agentic Post-training</b> , MCP, trajectory analysis, attribution analysis                                    |
| Data & Infrastructure | Simulation-based data generation, <b>Docker</b> , Docker Compose, <b>Slurm</b> , Git                             |
| Rendering             | <b>Unity</b> , <b>Blender</b> , MeshLab                                                                          |
| Languages             | Mandarin (native), English (TOEFL 115), Italian (entry-level)                                                    |